

6	A	Component of the weight parallel to the slope = $W \sin 30^\circ$ $= (1.0 \times 10^3)(0.50) = 0.50 \times 10^3 \text{ N}$ Because the barrel is moving at constant v, the applied $F = W \sin 30^\circ$ $WD = Fx = (0.50 \times 10^3)(5.0) = 2500 \text{ J}$

2

3

D

Useful WD in lifting the weight = $(480)(3.5) = 1680 \text{ J}$